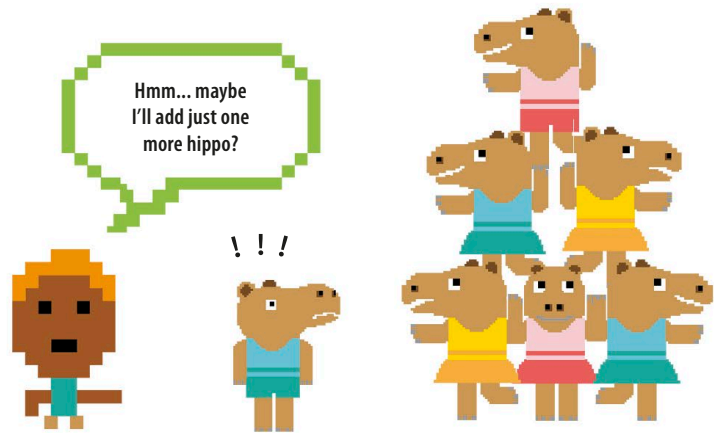


▷ How it works

The loop condition in Ahmed's program is `answer == 'y'`. This means that the user wants to add a hippo. In the body of the loop we add one to the number of hippos balanced, then ask the user if they want to add another. If they answer by typing "y" (for yes), the loop condition is True so we go round the loop again. If they answer "n" (no), the loop condition is False and the program leaves the loop.



Infinite loops

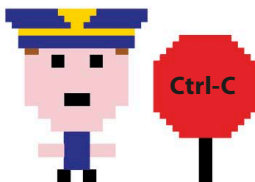
Sometimes you may want a **while** loop to keep going for as long as the program is running. This kind of loop is called an infinite loop. Lots of video-game programs use an infinite loop known as a main loop.



EXPERT TIPS

Stopping the loop

If you don't want an infinite loop, it's important to make sure that the body of a **while** loop does something that could make the loop condition False. But don't worry – if you accidentally code an infinite loop, you can escape from it by pressing the C key while holding down the Ctrl (control) key. You may have to press Ctrl-C several times before you quit the loop.



There is no False option to escape the loop.

```
>>> while True:
    print('This is an infinite loop!')
```

△ Into infinity

You make an infinite loop by setting the loop condition to a constant value: True. Because this value never changes, the loop will never exit. Try this **while** loop in the shell. It has no False option, so the loop will print "This is an infinite loop!" nonstop until you quit the program.

▽ Escaping infinity

You can deliberately use an infinite loop to get input from the user. This (annoying) program asks if the user is bored. As long as they type "n", it keeps asking the question. If they get fed up and type "y", it tells them they're rude and uses the **break** command to leave the loop!

The True condition is that the user is not bored yet ('n').

```
>>> while True:
    answer = input('Are you bored yet? (y/n)')
    if answer == 'y':
        print('How rude!')
        break
```

The False condition ('y') triggers the **break** command.